Natural Language Processing
Lab 3

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16-17 September 2026

Today's Agenda

1. Terminology
2. Vector Representations
    a. One-Hot Encoding
    b. Bag of Words
    c. Bag of N-Grams
    d. TF-IDF (Math)
3. Cosine Similarity (Math)
4. Make and Test a Search Engine! (Jupyter Notebook)

Terminology

'Algorithmic bias describes systematic and repeatable errors in a ...'

‘Bias can emerge due to many factors, including but not limited ...’

‘Algorithmic bias is found across platforms, including but not ...’

‘The study of algorithmic bias is most concerned with algorithms ...’

Terminology

corpus

'Algorithmic bias describes systematic and repeatable errors in a ...'

‘Bias can emerge due to many factors, including but not limited ...’

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‘The study of algorithmic bias is most concerned with algorithms ...’

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Terminology

document

'Algorithmic bias describes systematic and repeatable errors in a ...'

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Terminology

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Terminology

token, term

'Algorithmic bias describes systematic and repeatable errors in a ...'

‘Bias can emerge due to many factors, including but not limited ...’

‘Algorithmic bias is found across platforms, including but not ...’

‘The study of algorithmic bias is most concerned with algorithms ...’

Terminology

token, term → word, N-gram, subword, …

'Algorithmic bias describes systematic and repeatable errors in a ...'

‘Bias can emerge due to many factors, including but not limited ...’

‘Algorithmic bias is found across platforms, including but not ...’

‘The study of algorithmic bias is most concerned with algorithms ...’

Terminology

doc1

doc2

doc3

doc4

across	algorithm	algorithmic	and	arbitrary	bias	...
int/float	int/float	int/float	int/float	int/float	int/float	int/float

Terminology

doc1

doc2

doc3

doc4

vocabulary, lexicon (ordered)

across	**algorithm**	**algorithmic**	**and**	**arbitrary**	**bias**	**...**
int/float	int/float	int/float	int/float	int/float	int/float	int/float

Terminology

doc1

doc2

doc3

doc4

vocabulary, lexicon (ordered) → feature, dimesnion

across	**algorithm**	**algorithmic**	**and**	**arbitrary**	**bias**	**...**
int/float	int/float	int/float	int/float	int/float	int/float	int/float

Terminology

doc1

doc2

doc3

doc4

across	algorithm	algorithmic	and	arbitrary	bias	...
int/float	int/float	int/float	int/float	int/float	int/float	int/float

Terminology

doc1

doc2

doc3

doc4

across	algorithm	algorithmic	and	arbitrary	bias	...
int/float	int/float	int/float	int/float	int/float	int/float	int/float

vector

Terminology

doc1

doc2

doc3

doc4

across	algorithm	algorithmic	and	arbitrary	bias	...
int/float	int/float	int/float	int/float	int/float	int/float	int/float

Vector Space Model (VSM)

One-Hot Encoding

'Algorithmic bias describes systematic and repeatable errors in a ...'

	across	algorithm	algorithmic	and	arbitrary	bias	...
algorithmic	0	0	1	0	0	0	
bias	0	0	0	0	0	1	
describes	0	0	0	0	0	0	
systematic	0	0	0	0	0	0	
and	0	0	0	1	0	0	

One-Hot Encoding

✅ retain sequence
❌ super sparse

'Algorithmic bias describes systematic and repeatable errors in a ...'

	across	algorithm	algorithmic	and	arbitrary	bias	...
algorithmic	0	0	1	0	0	0	
bias	0	0	0	0	0	1	
describes	0	0	0	0	0	0	
systematic	0	0	0	0	0	0	
and	0	0	0	1	0	0	

Bag of Words

doc1

doc2

doc3

doc4

	across	algorithm	algorithmic	and	arbitrary	bias	...
doc 1	0	0	1	1	1	1	
doc 2	0	2	0	0	0	1	
doc 3	1	0	1	3	0	1	
doc 4	0	0	1	1	0	1	

Bag of Words

✅ squeeze all docs
❌ lose sequence

doc1 doc2

doc3 doc4

	across	algorithm	algorithmic	and	arbitrary	bias	...
doc 1	0	0	1	1	1	1	
doc 2	0	2	0	0	0	1	
doc 3	1	0	1	3	0	1	
doc 4	0	0	1	1	0	1	

Bag of Words

✅ squeeze all docs
❌ lose sequence

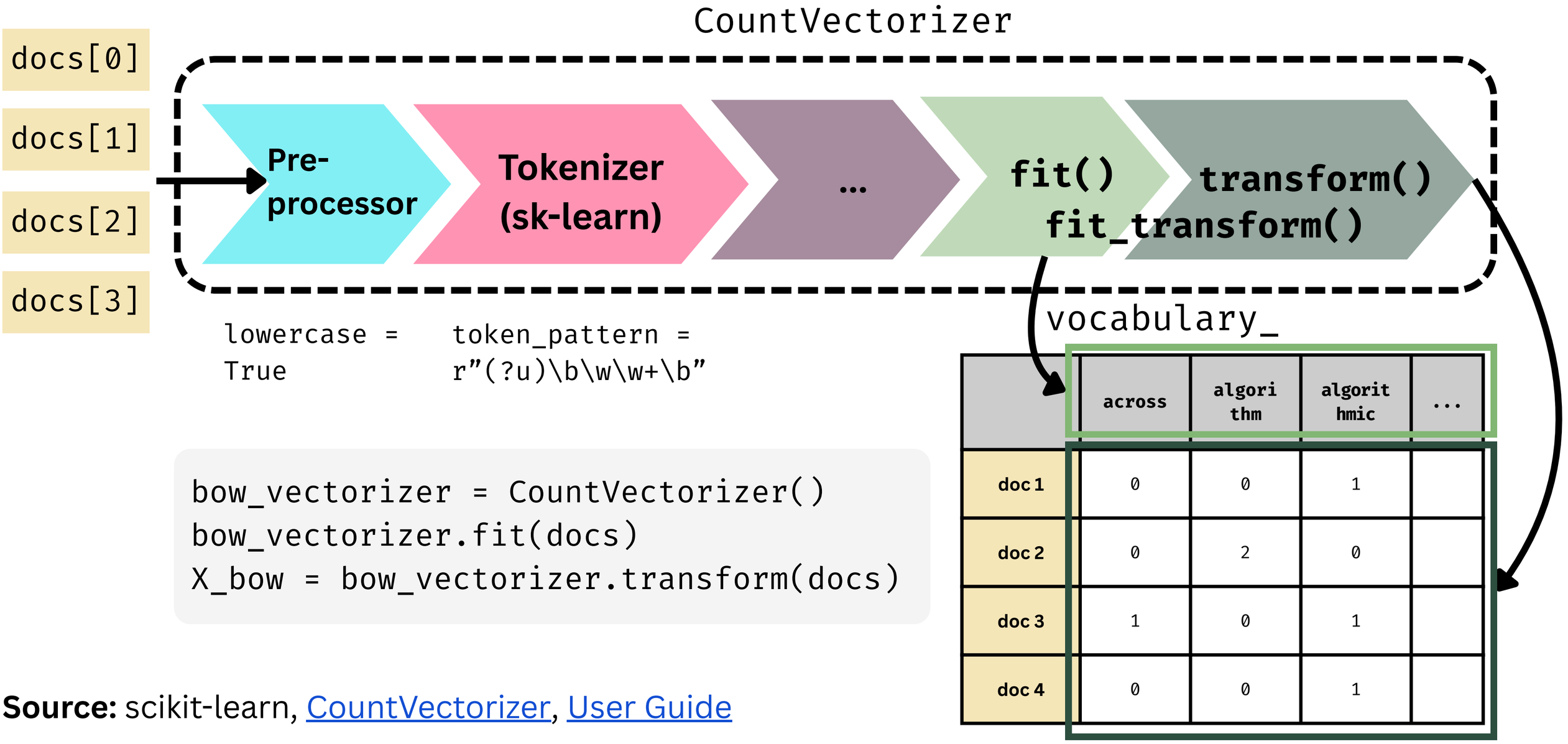
doc1 doc2

doc3 doc4

document-term matrix

	across	algorithm	algorithmic	and	arbitrary	bias	...
doc 1	0	0	1	1	1	1	
doc 2	0	2	0	0	0	1	
doc 3	1	0	1	3	0	1	
doc 4	0	0	1	1	0	1	

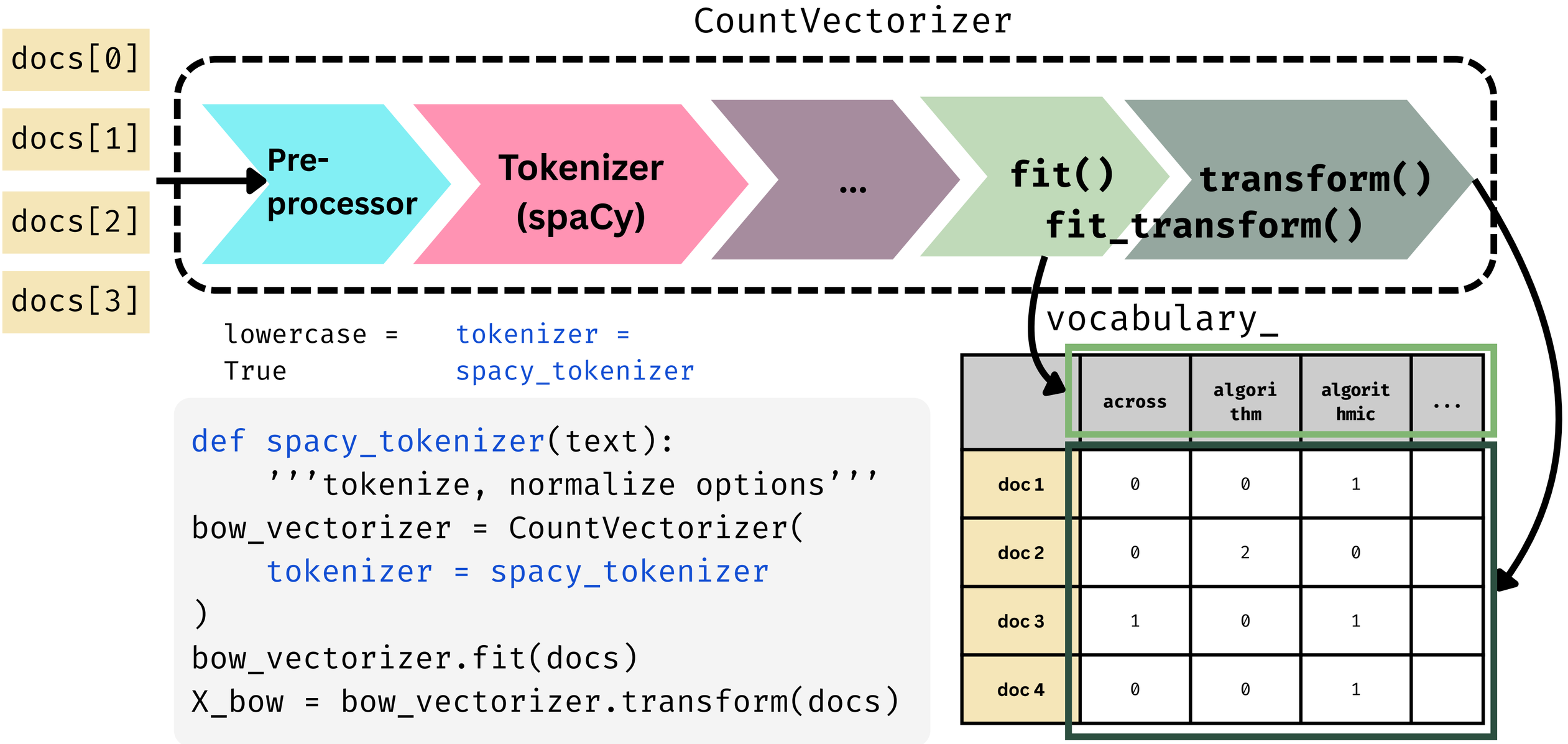
Bag of Words: CountVectorizer (default)




```
bow_vectorizer = CountVectorizer()
bow_vectorizer.fit(docs)
X_bow = bow_vectorizer.transform(docs)
```

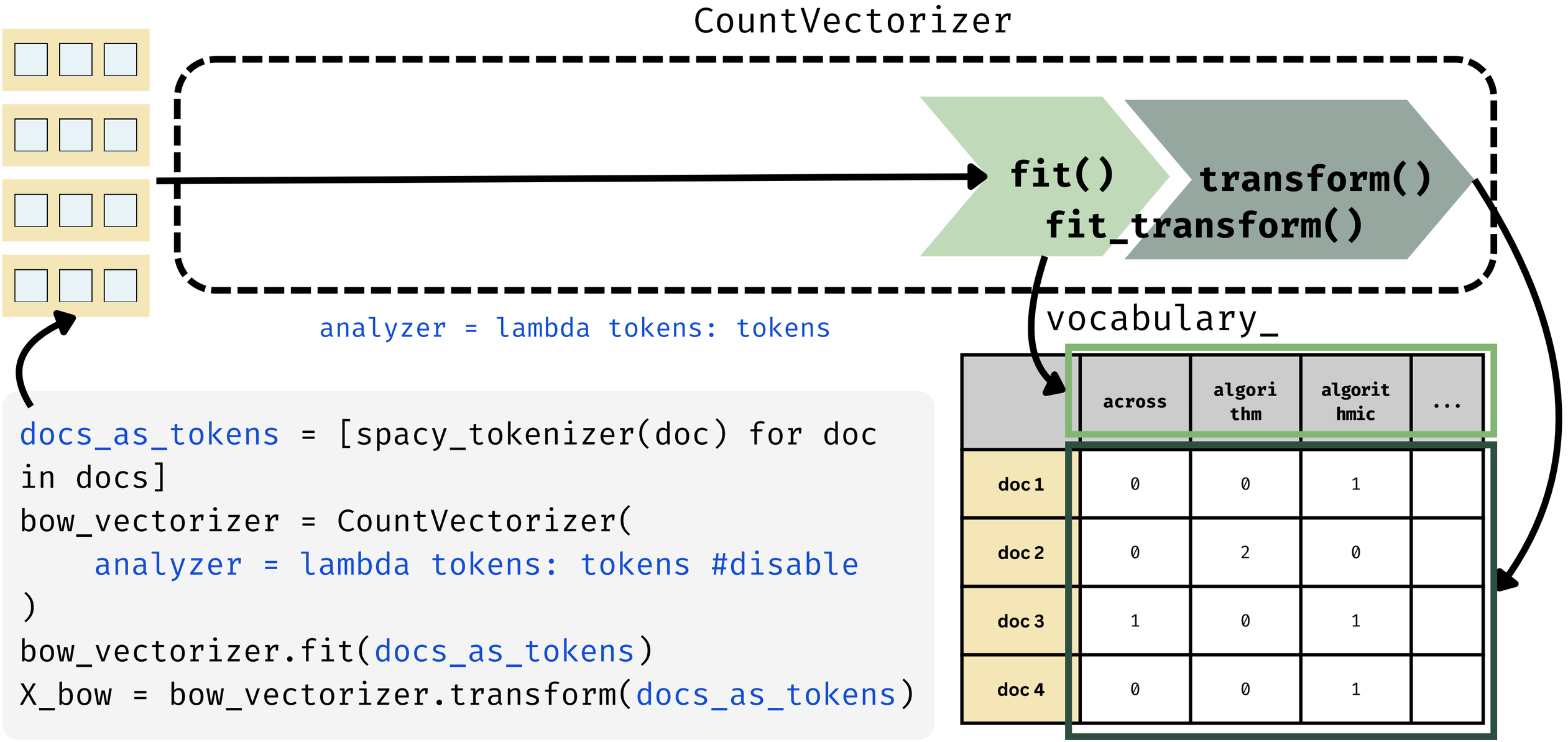
Source: scikit-learn, CountVectorizer, User Guide

Bag of Words: CountVectorizer (+spaCy)




```
def spacy_tokenizer(text):
    '''tokenize, normalize options'''
bow_vectorizer = CountVectorizer(
    tokenizer = spacy_tokenizer
)
bow_vectorizer.fit(docs)
X_bow = bow_vectorizer.transform(docs)
```

BoW: CountVectorizer (external tokenize)




```
docs_as_tokens = [spacy_tokenizer(doc) for doc
in docs]
bow_vectorizer = CountVectorizer(
    analyzer = lambda tokens: tokens #disable
)
bow_vectorizer.fit(docs_as_tokens)
X_bow = bow_vectorizer.transform(docs_as_tokens)
```

Bag of N-Grams

doc1

doc2

doc3

doc4

	across	across platforms	algorithm	algorithm or	algorithmic	algorithmic bias	...
doc 1	0	0	0	0	1	1	
doc 2	0	0	2	1	0	0	
doc 3	1	1	0	0	1	1	
doc 4	0	0	0	0	1	1	

Bag of N-Grams

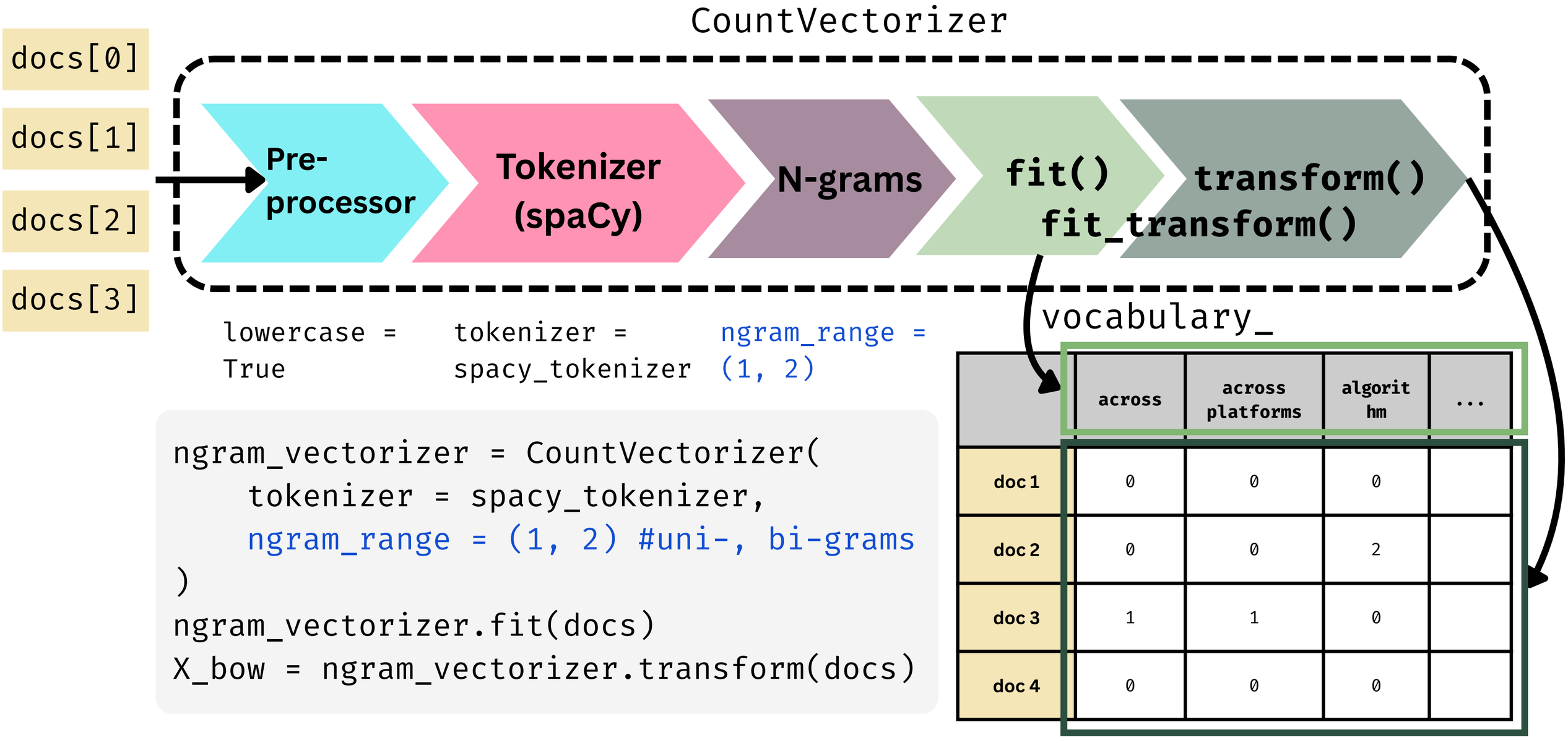
✅ neighboring sequence
❌ vocabulary size increase

doc1 doc2

doc3 doc4

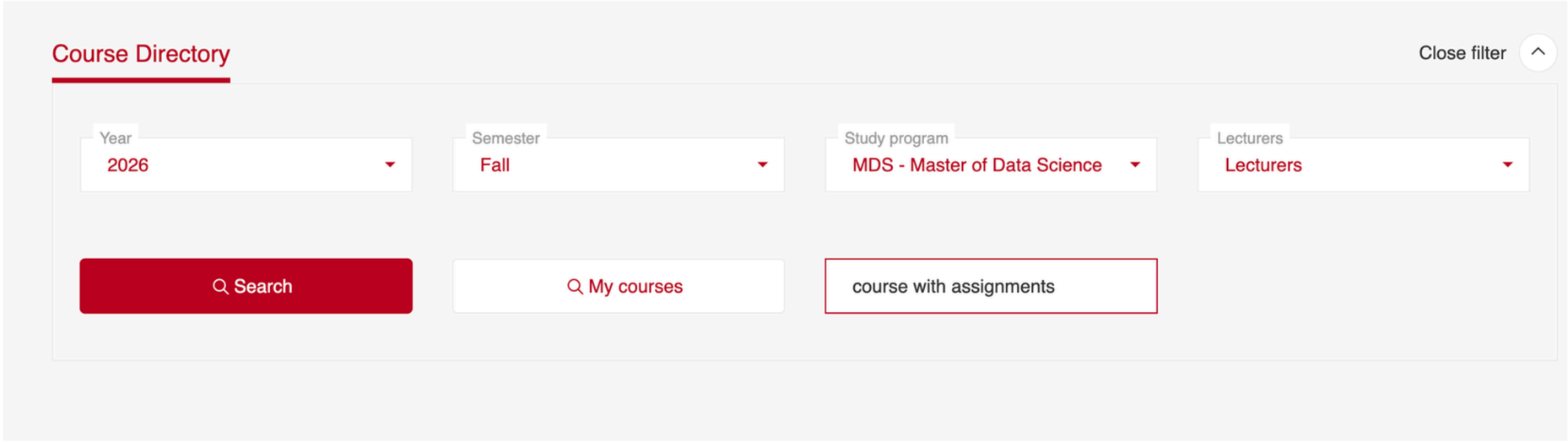
	across	across platforms	algorithm	algorithm or	algorithmic	algorithmic bias	...
doc 1	0	0	0	0	1	1	
doc 2	0	0	2	1	0	0	
doc 3	1	1	0	0	1	1	
doc 4	0	0	0	0	1	1	

Bag of N-Grams: CountVectorizer (+spaCy)

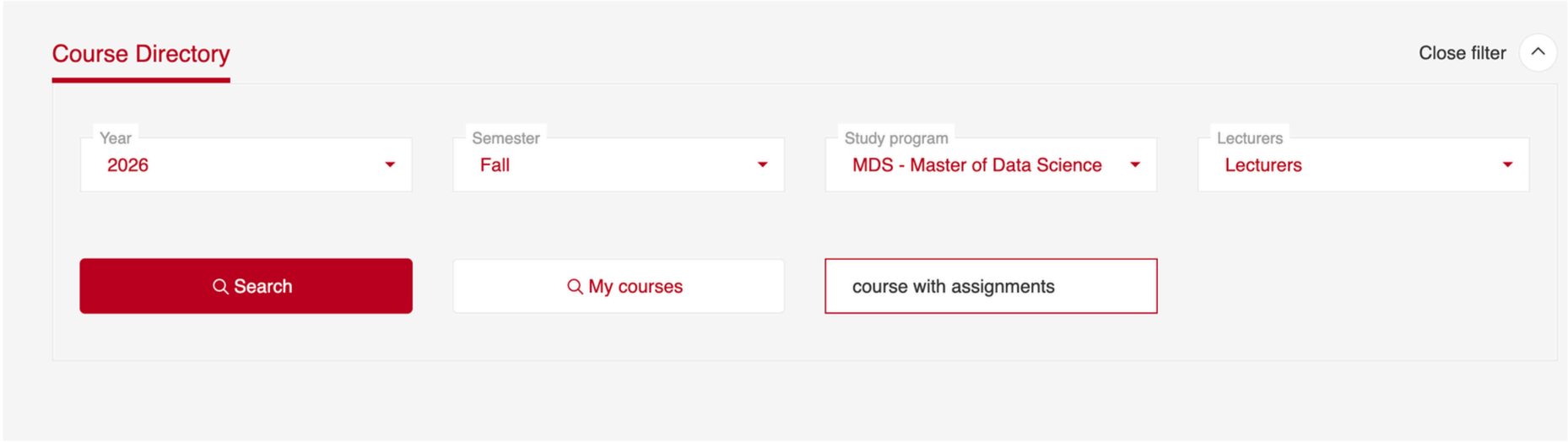


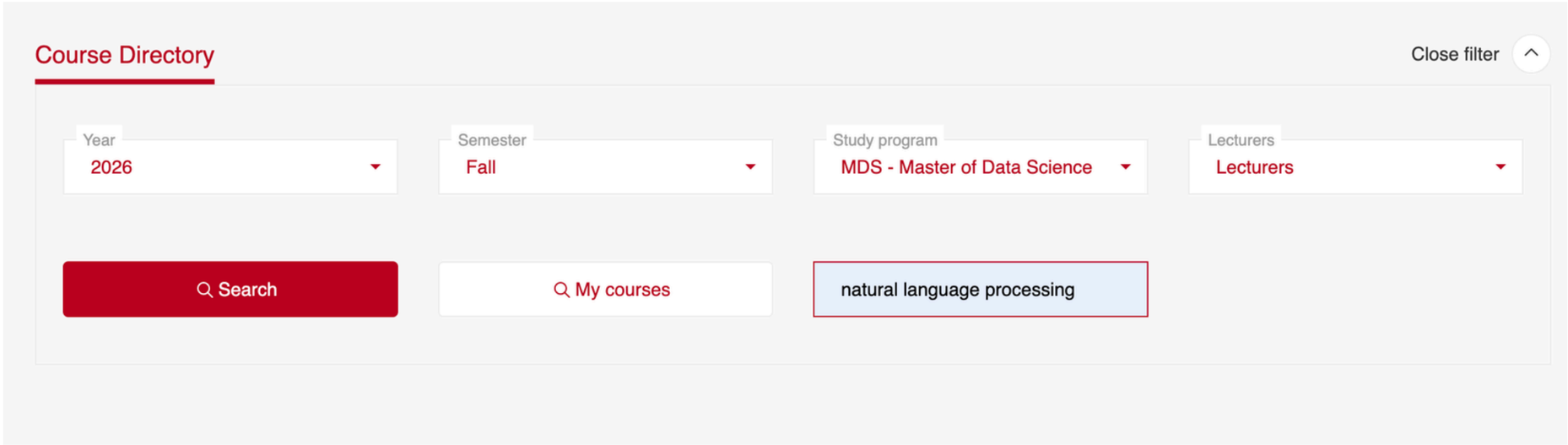

```
ngram_vectorizer = CountVectorizer(
    tokenizer = spacy_tokenizer,
    ngram_range = (1, 2) #uni-, bi-grams
)
ngram_vectorizer.fit(docs)
X_bow = ngram_vectorizer.transform(docs)
```

TF-IDF: Intuition



TF-IDF: Intuition





TF-IDF: Math

1. **Term Frequency**: how often does the term appear in THIS document?
2. **Inverse Document Frequency**: how rare is the term relative to the entire corpus?
    a. N_d: total number of documents
    b. DF(t): number of documents the term appears in

word	TF(t, d)	N_d	DF(t)	IDF	TF*IDF
or	1	4	4		
biases	1	4	1		
platforms	2	4	1		

TF-IDF: Math

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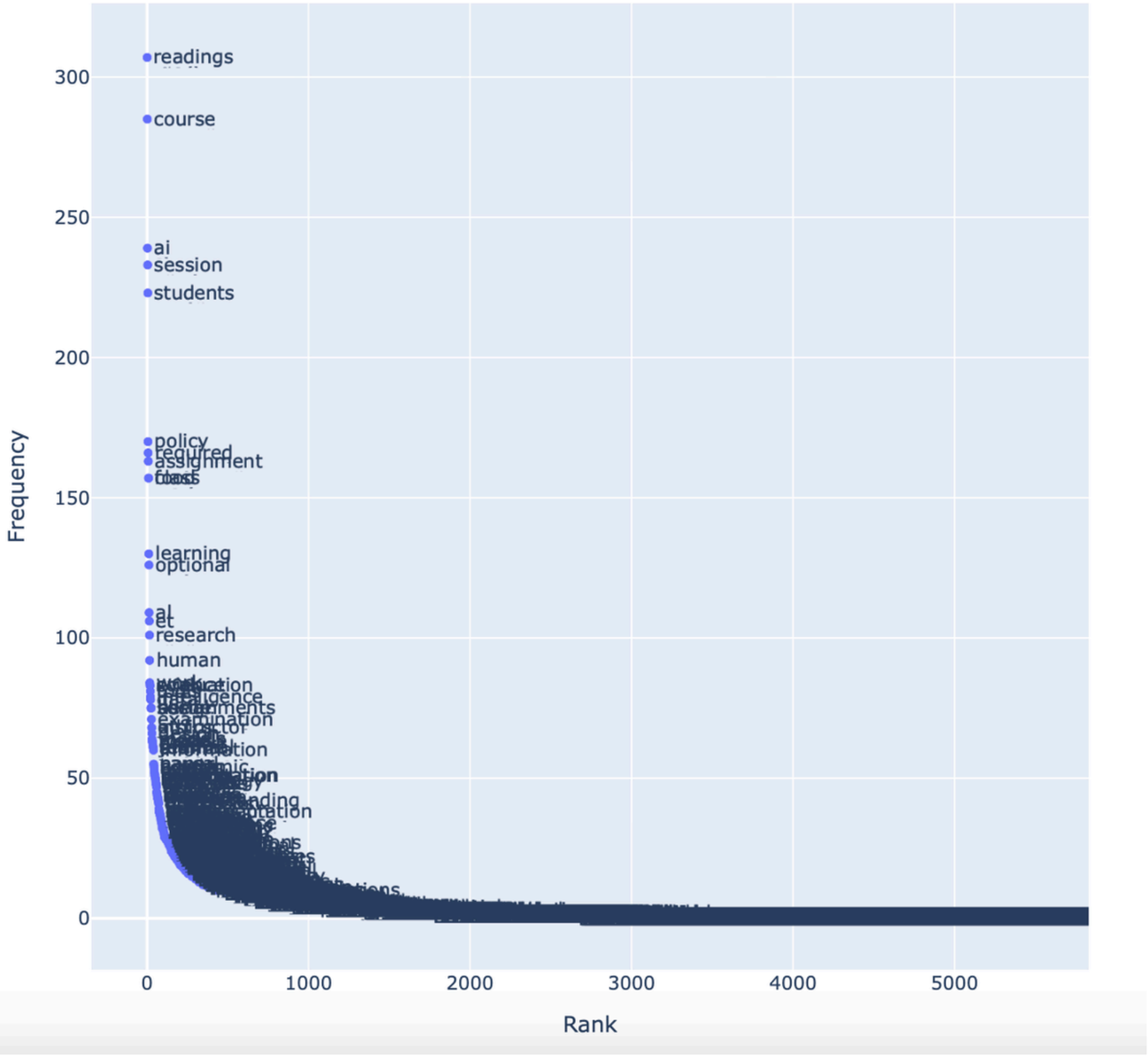
word	TF(t, d)	N_d	DF(t)	IDF	TF*IDF
or	1	4	4	4/4	
biases	1	4	1	4/1	
platforms	2	4	1	4/1	

TF-IDF: Math

1. **Term Frequency**: how often does the term appear in THIS document?
2. **Inverse Document Frequency**: how rare is the term relative to the entire corpus?
    a. N_d: total number of documents
    b. DF(t): number of documents the term appears in

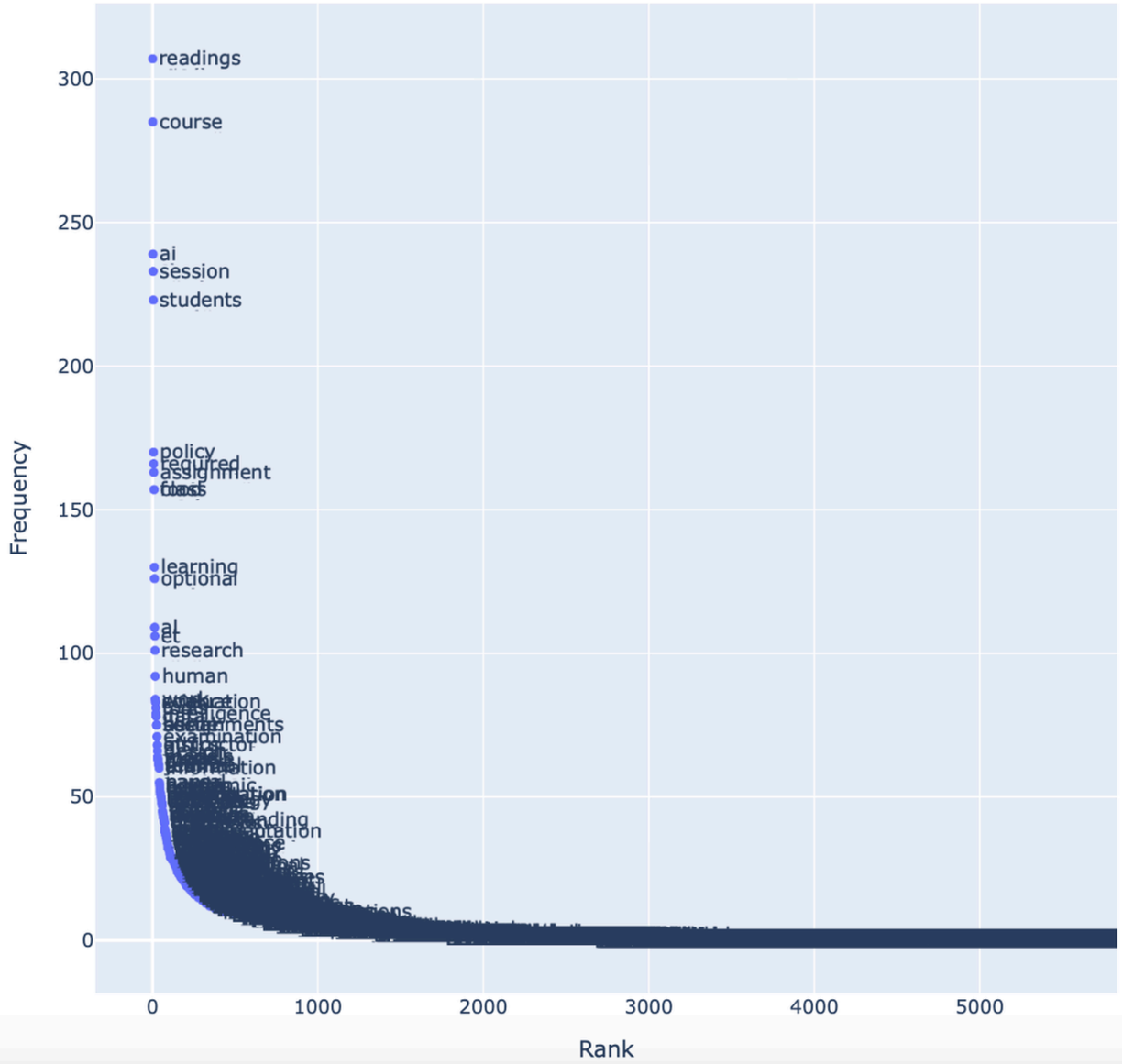
word	TF(t, d)	N_d	DF(t)	IDF	TF*IDF
or	1	4	4	4/4	1
biases	1	4	1	4/1	4
platforms	2	4	1	4/1	8

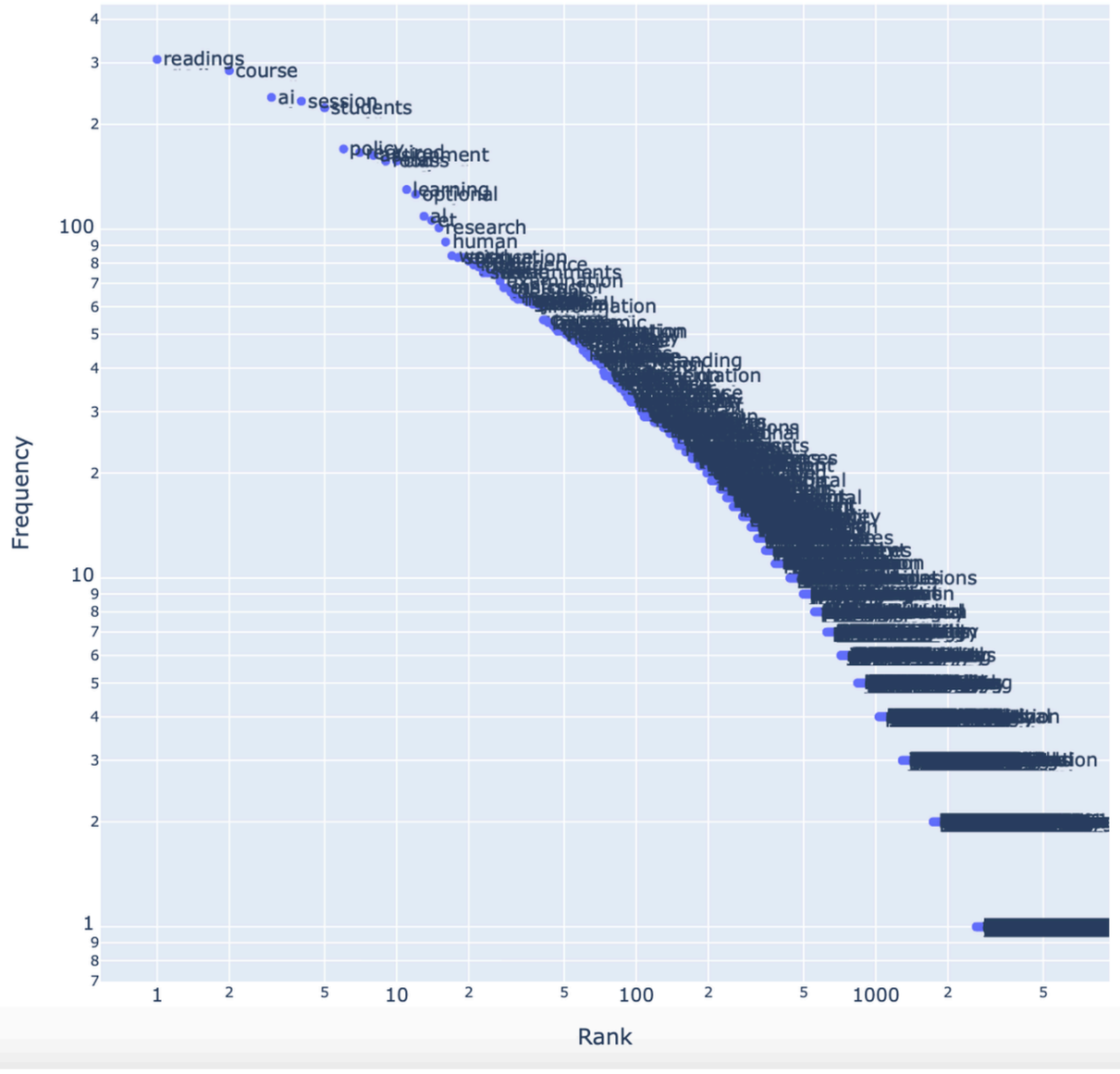
TF-IDF and Zipf’s Law



TF-IDF and Zipf's Law

log-log plot





TF-IDF: Math

$$IDF = \ln\left(\frac{N_d}{DF(t)}\right)$$

word	TF(t, d)	N_d	DF(t)	IDF	TF*IDF
or	1	4	4	0	0
biases	1	4	1	1.39	1.39
platforms	2	4	1	1.39	2.77

TF-IDF: Math

Source: scikit-learn, TfidfTransformer

$$IDF = \ln\left(\frac{1+N_d}{1+DF(t)}\right) + 1$$

word	TF(t, d)	N_d	DF(t)	IDF	TF*IDF
or	1	4	4	1	1
biases	1	4	1	1.92	1.92
platforms	2	4	1	1.92	3.83

TF-IDF: Math

Source: scikit-learn, TfidfTransformer

$$IDF = \ln\left(\frac{1+N_d}{1+DF(t)}\right) + 1$$

avoid division by zero
for OOV terms

word	TF(t, d)	N_d	DF(t)	IDF	TF*IDF
or	1	4	4	1	1
biases	1	4	1	1.92	1.92
platforms	2	4	1	1.92	3.83

TF-IDF: Math

Source: scikit-learn, [TfidfTransformer]

$$IDF = \ln\left(\frac{1+N_d}{1+DF(t)}\right) + 1$$

avoid over-penalizing very common terms

word	TF(t, d)	N_d	DF(t)	IDF	TF*IDF
or	1	4	4	1	1
biases	1	4	1	1.92	1.92
platforms	2	4	1	1.92	3.83

TF-IDF

doc1 doc2

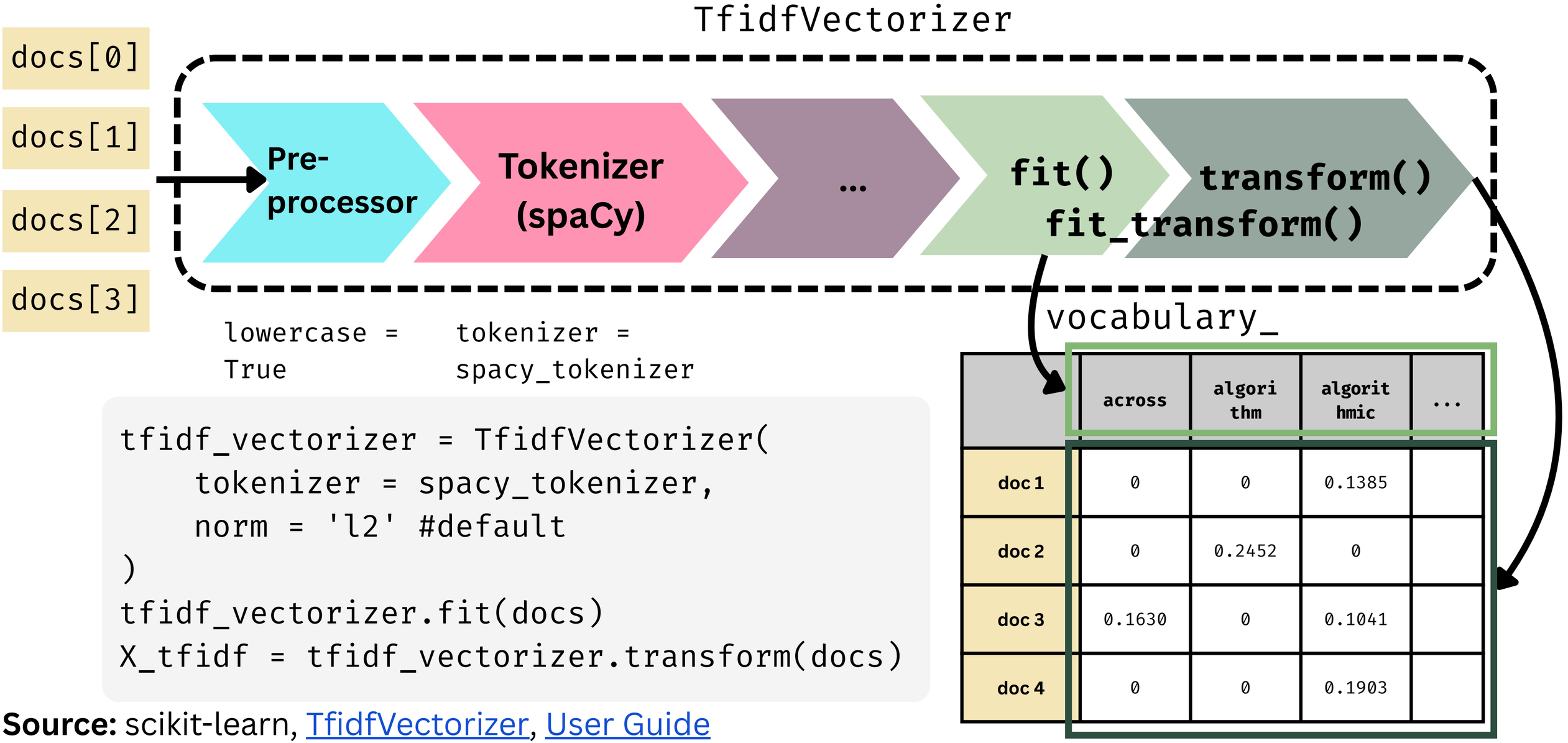
doc3 doc4

Source: scikit-learn, TfidfVectorizer

Also divides by L2 (Euclidean) norm of each vector by default

	across	algorithm	algorithmic	and	arbitrary	bias	...
doc 1	0	0	0.138548	0.138548	0.217062	0.113272	
doc 2	0	0.245172	0	0	0	0.063971	
doc 3	0.163038	0	0.104065	0.312194	0	0.085080	
doc 4	0	0	0.190273	0.190273	0	0.155561	

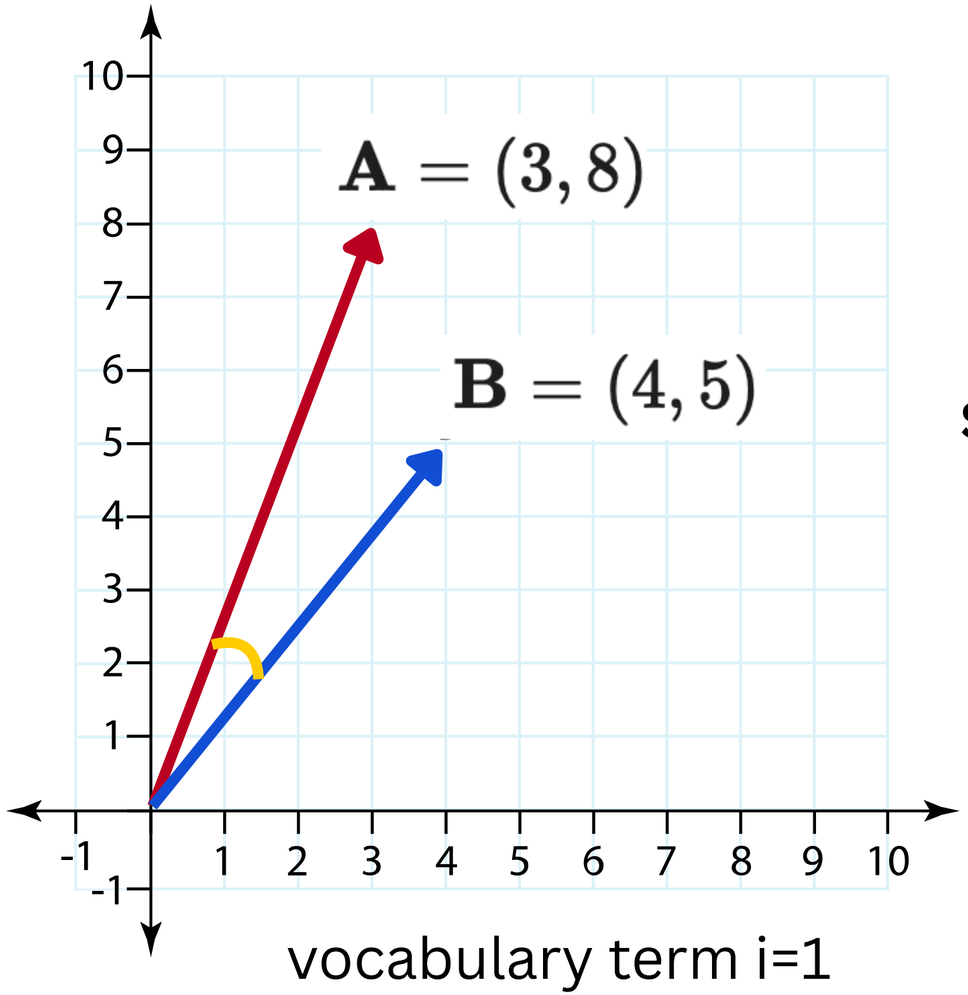
TF-IDF: TfidfVectorizer (+spaCy)




```
tfidf_vectorizer = TfidfVectorizer(
    tokenizer = spacy_tokenizer,
    norm = 'l2' #default
)
tfidf_vectorizer.fit(docs)
X_tfidf = tfidf_vectorizer.transform(docs)
```

Source: scikit-learn, TfidfVectorizer, User Guide

Cosine Similarity: Math

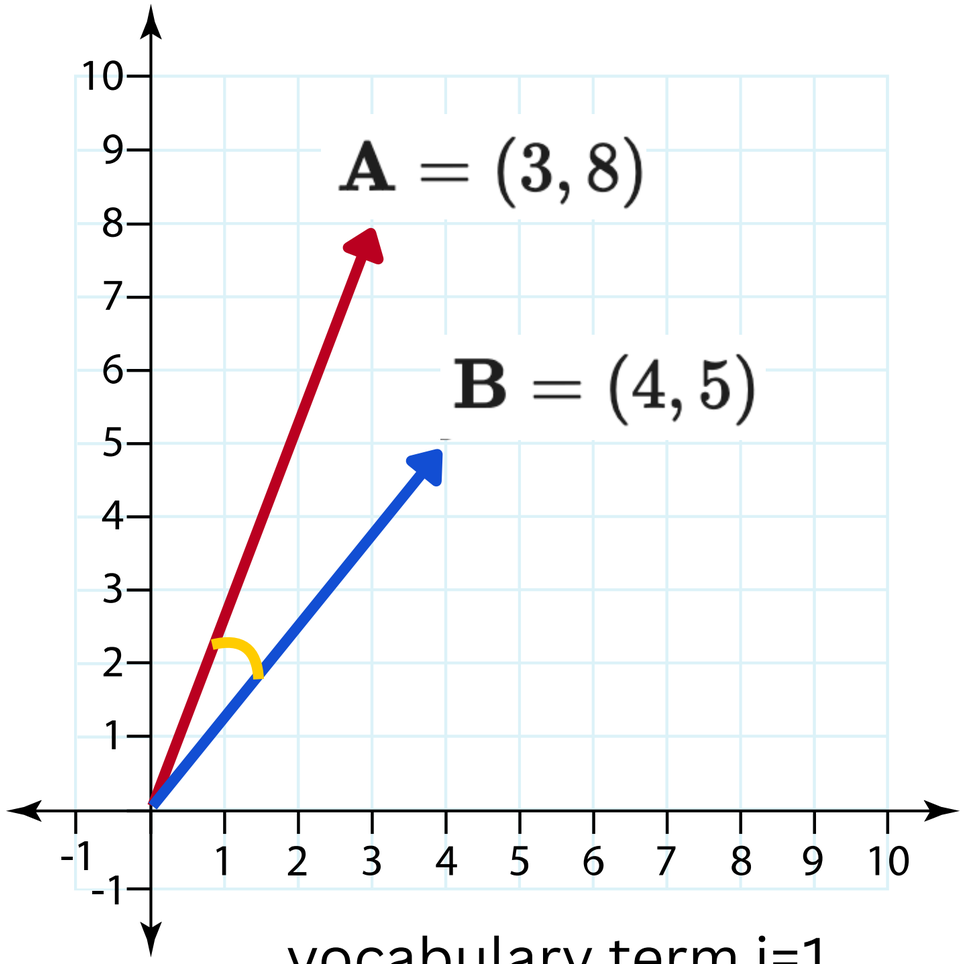


$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

Source: Vajjala et al. (2020, Ch. 3, p. 84)

Cosine Similarity: Math

vocabulary term i=2



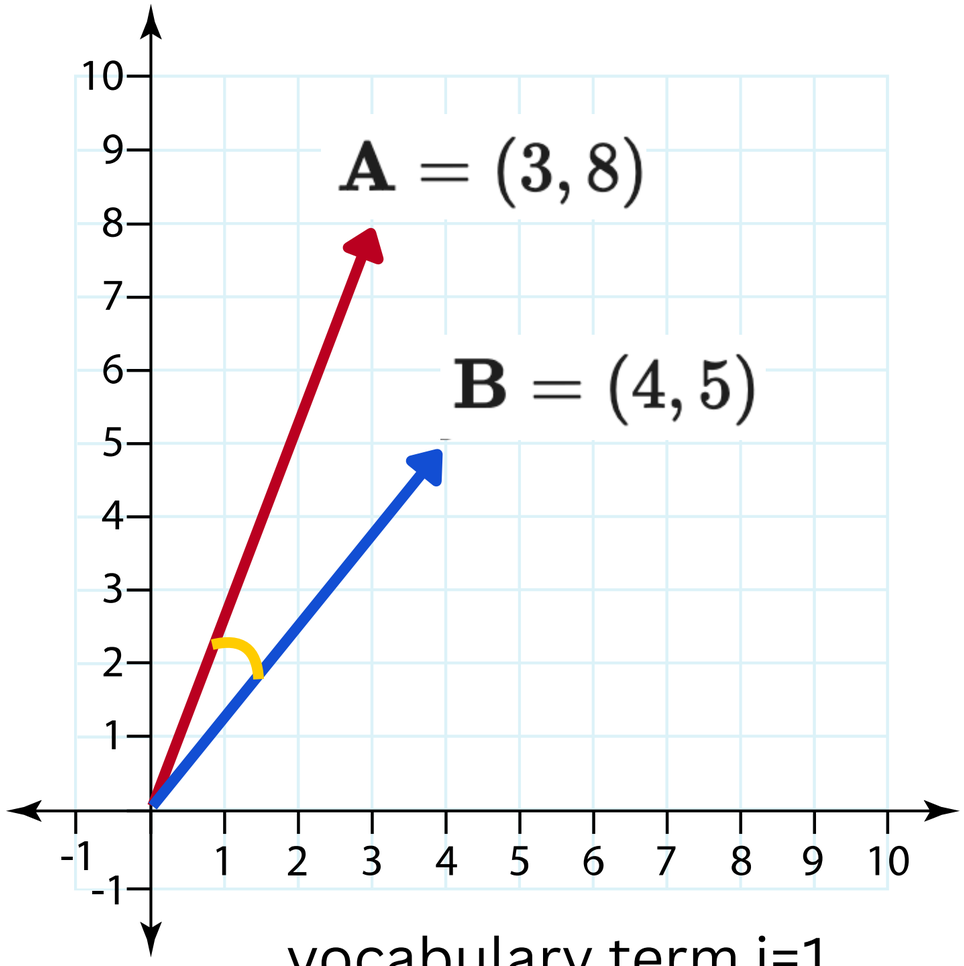
vocabulary term i=1

$$\mathbf{A} \cdot \mathbf{B} =$$

Dot product:
sum of pair-wise products

Cosine Similarity: Math

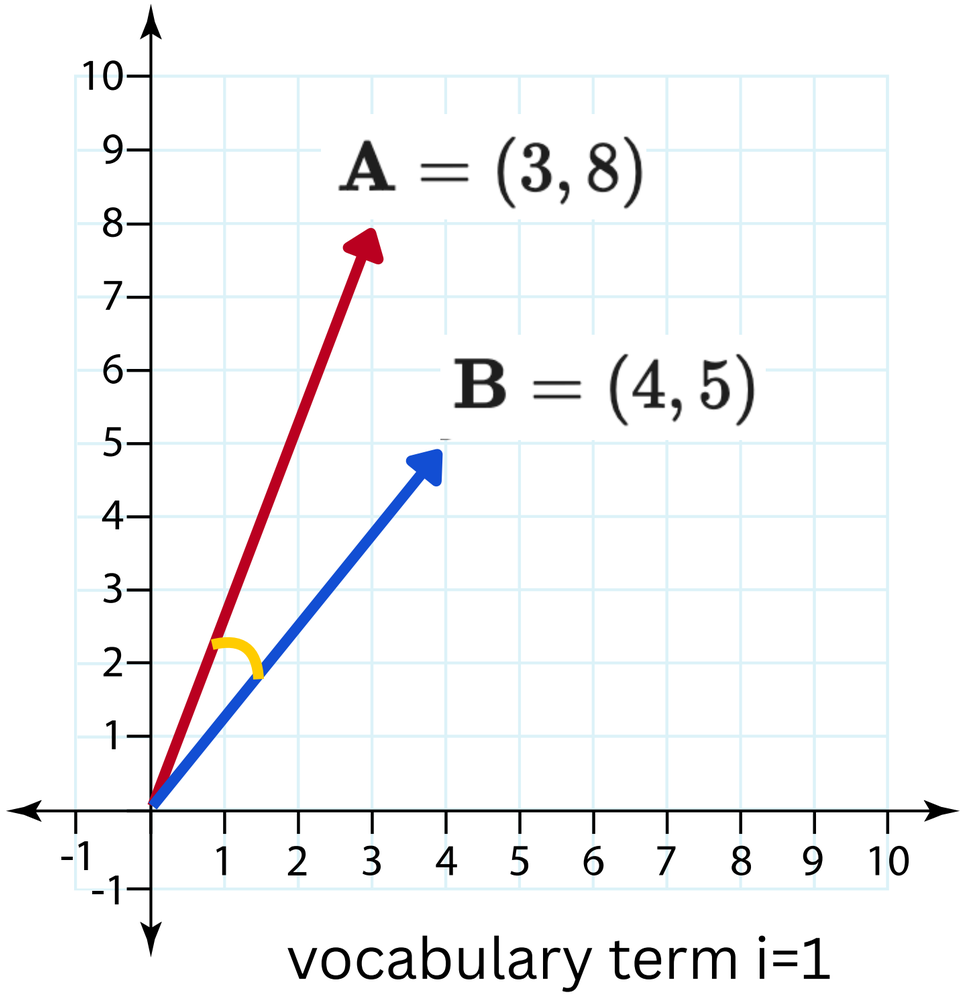
vocabulary term i=2



vocabulary term i=1

$$\mathbf{A} \cdot \mathbf{B} = \sum_{i=1}^{n} A_i B_i$$

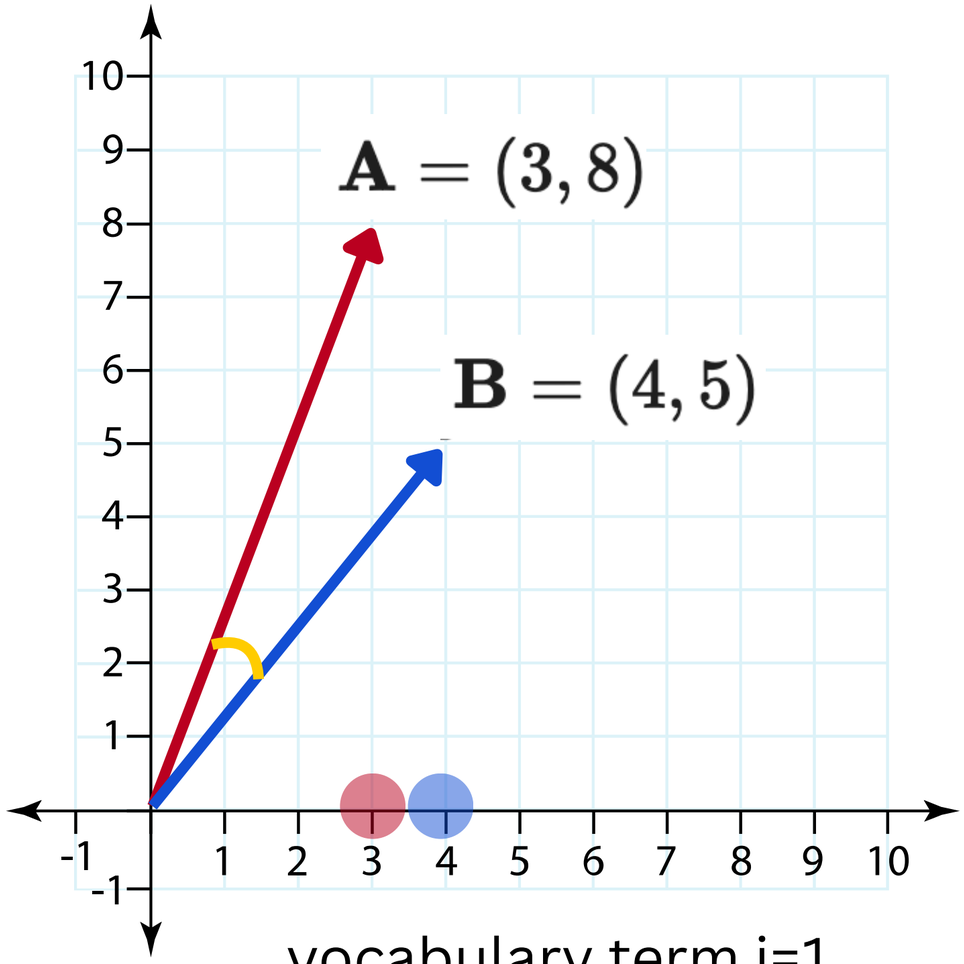
Cosine Similarity: Math



$$\mathbf{A} \cdot \mathbf{B} = A_1 B_1 + A_2 B_2$$

Cosine Similarity: Math

vocabulary term i=2

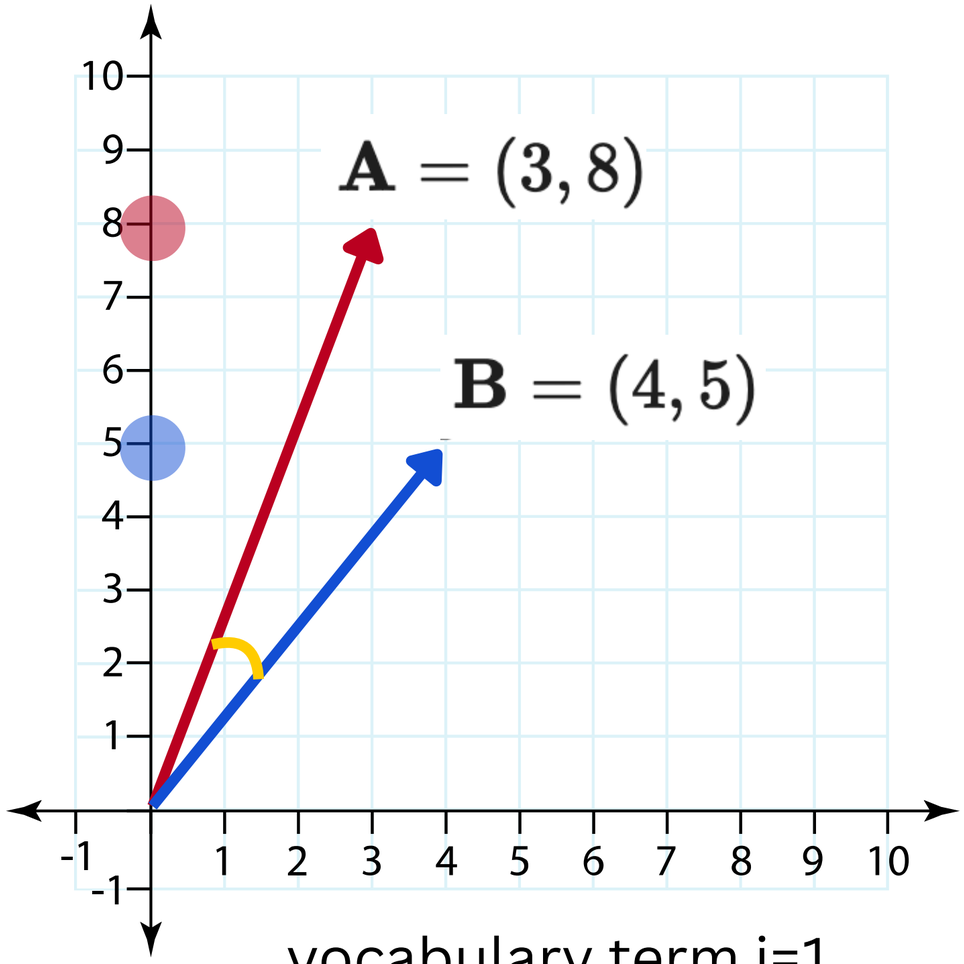


vocabulary term i=1

$$\mathbf{A} \cdot \mathbf{B} = 3 * 4 + A_2 B_2$$

Cosine Similarity: Math

vocabulary term i=2

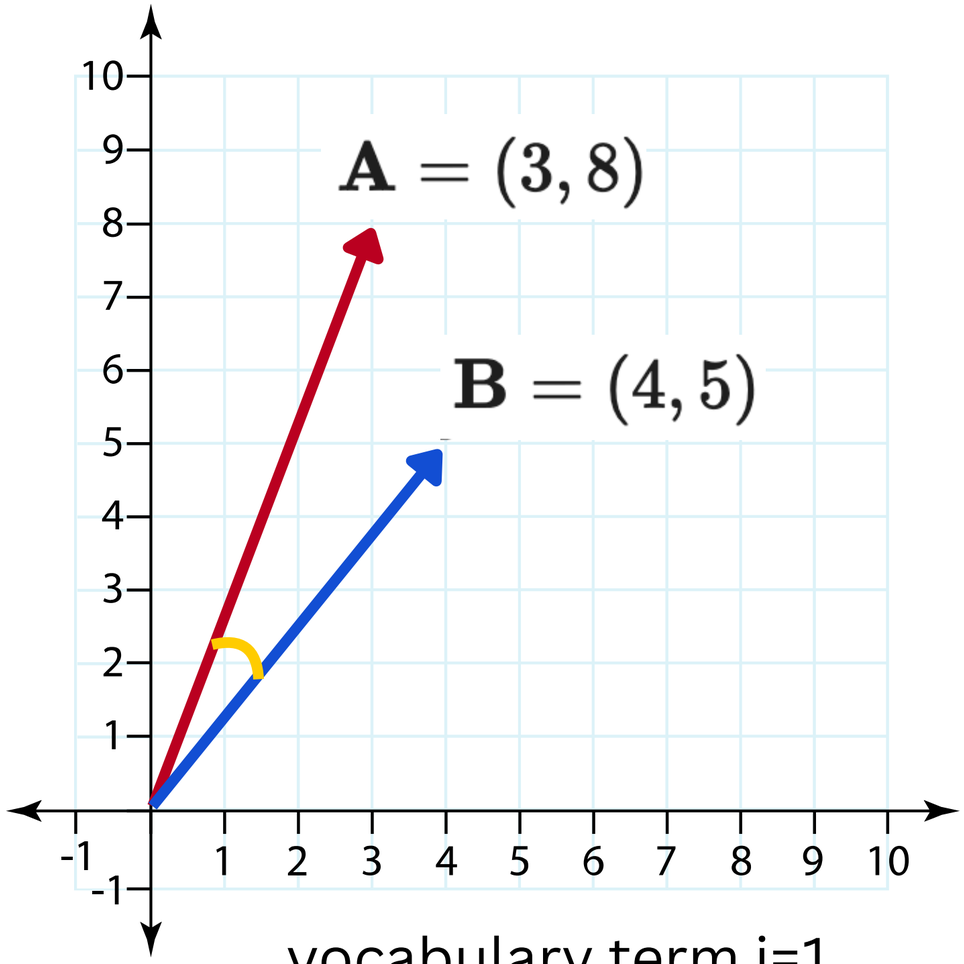


vocabulary term i=1

$$\mathbf{A} \cdot \mathbf{B} = 12 + 8 * 5$$

Cosine Similarity: Math

vocabulary term i=2

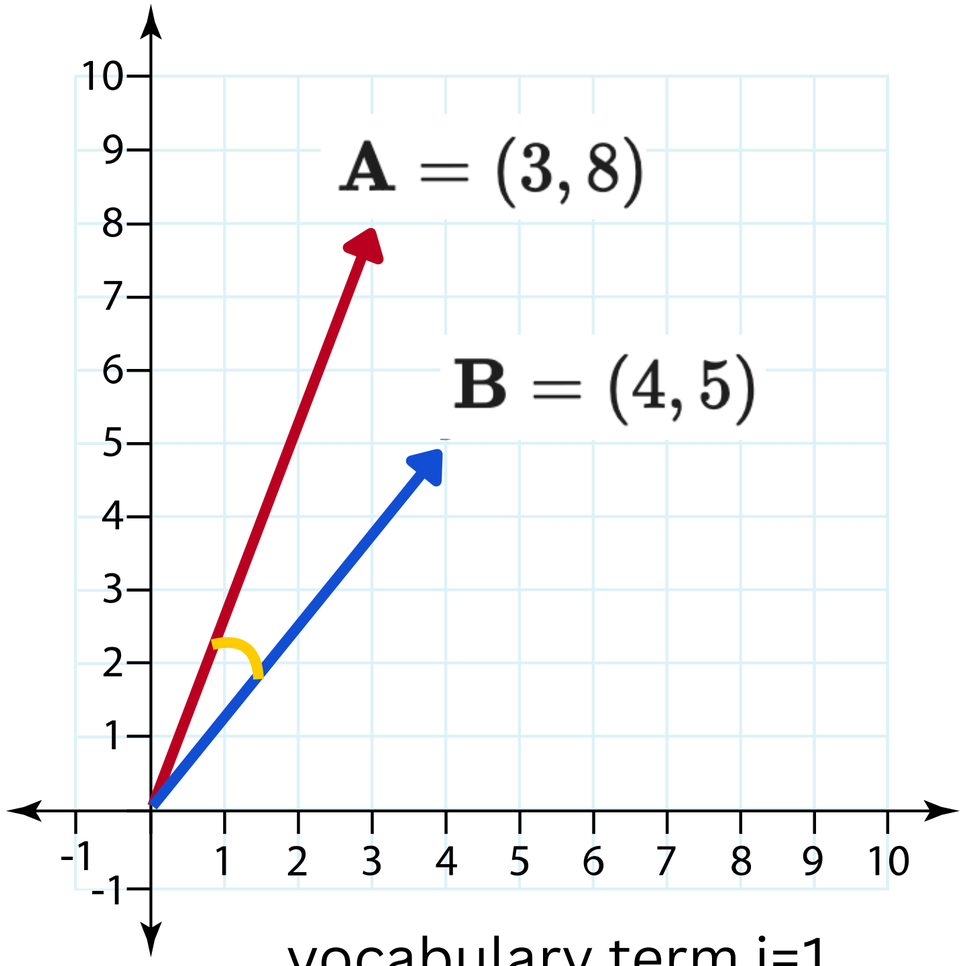


vocabulary term i=1

$$\mathbf{A} \cdot \mathbf{B} = 12 + 40$$

Cosine Similarity: Math

vocabulary term i=2

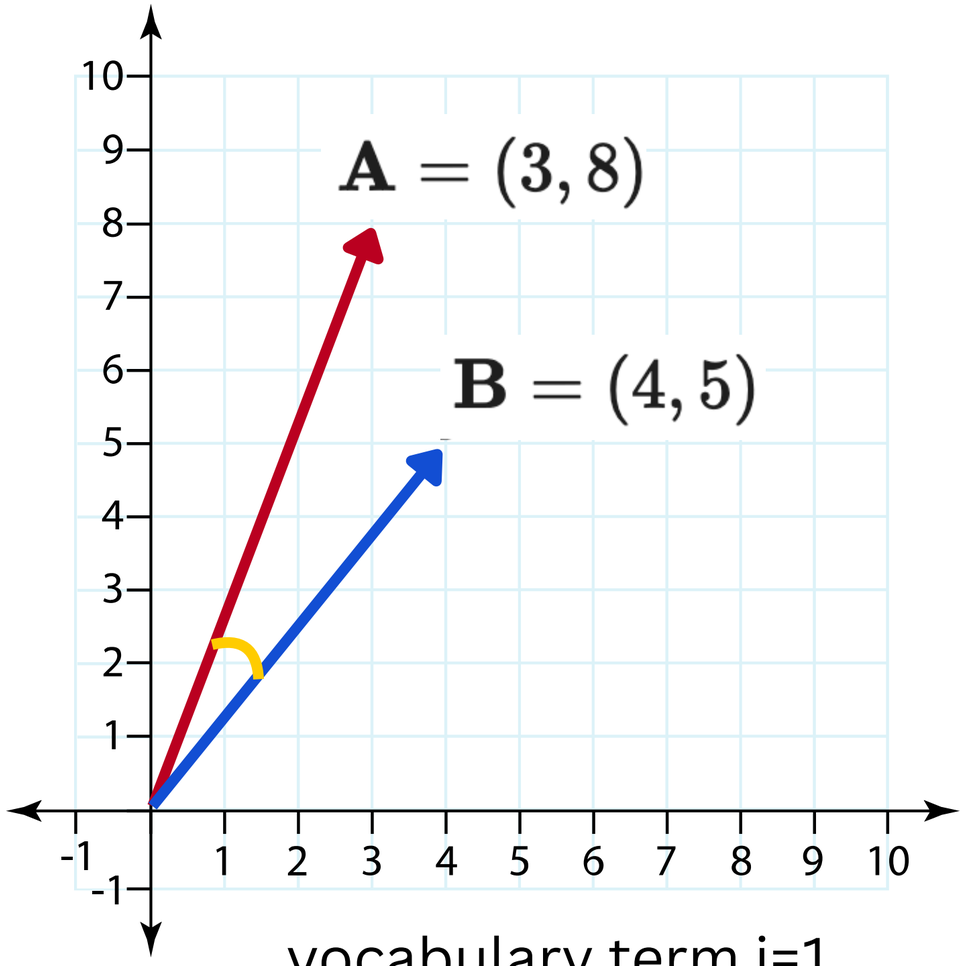


vocabulary term i=1

$$\mathbf{A} \cdot \mathbf{B} = 52$$

Cosine Similarity: Math

vocabulary term i=2

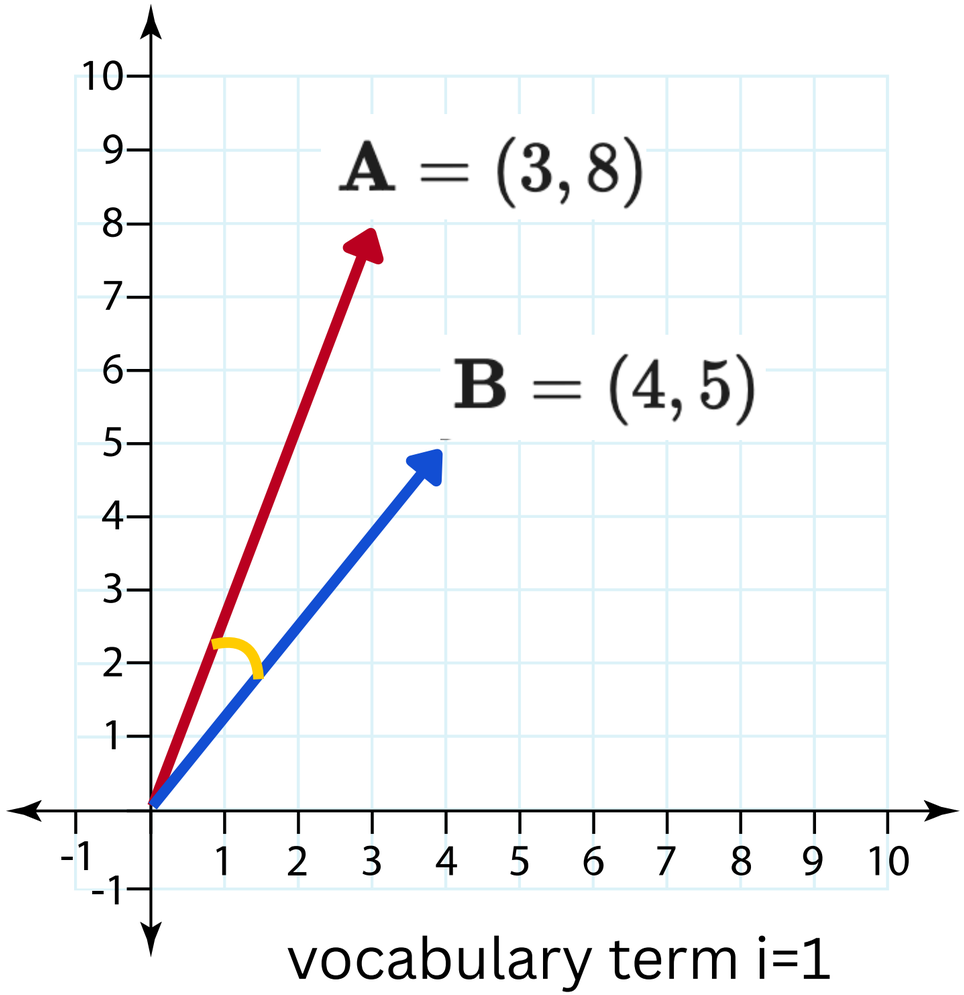


vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{52}{\text{Product of L2 (Euclidean) norms}}$$

Cosine Similarity: Math

vocabulary term i=2

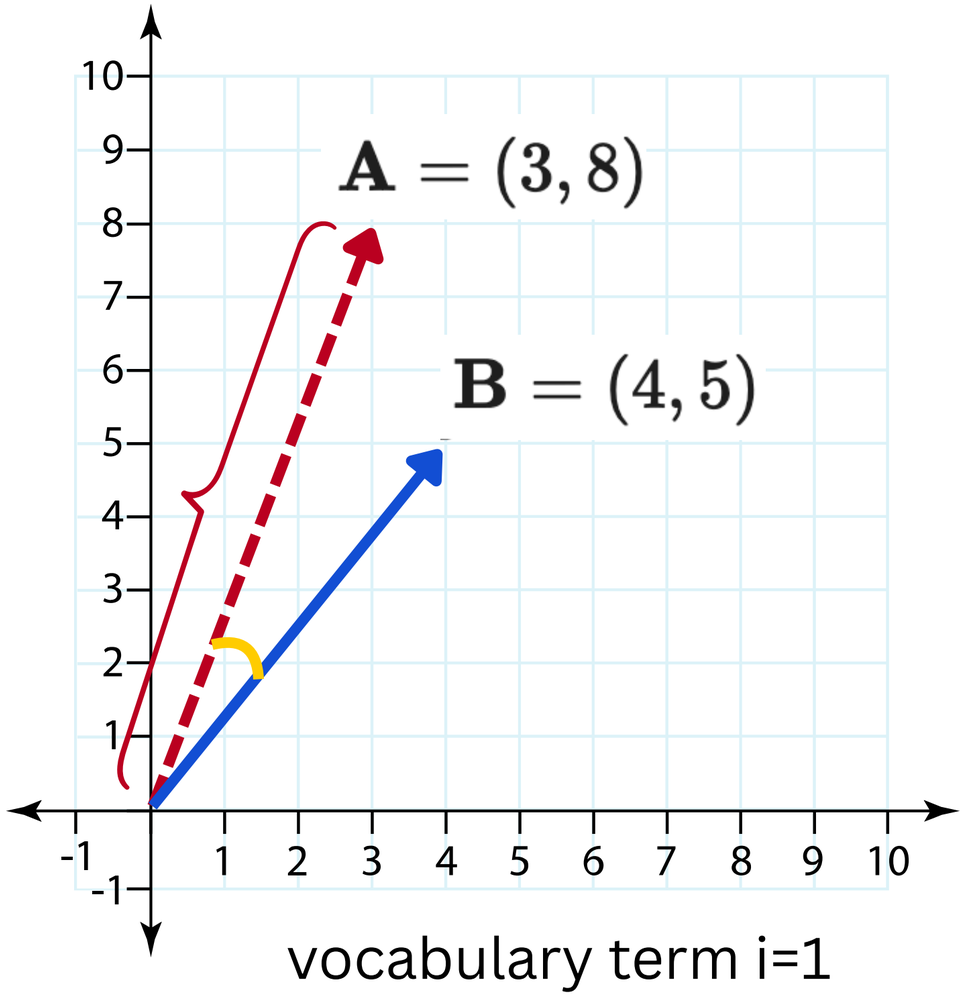


vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{52}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}$$

Cosine Similarity: Math

vocabulary term i=2

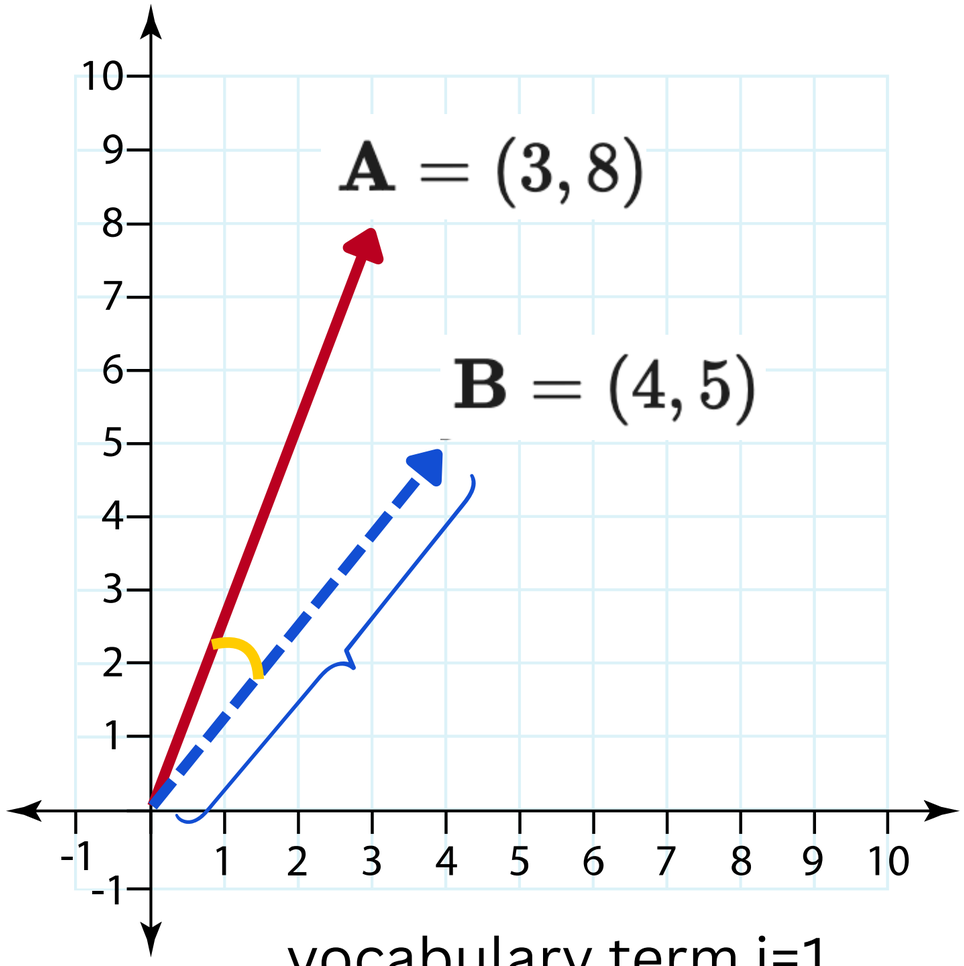


vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{52}{\sqrt{3^2 + 8^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

Cosine Similarity: Math

vocabulary term i=2

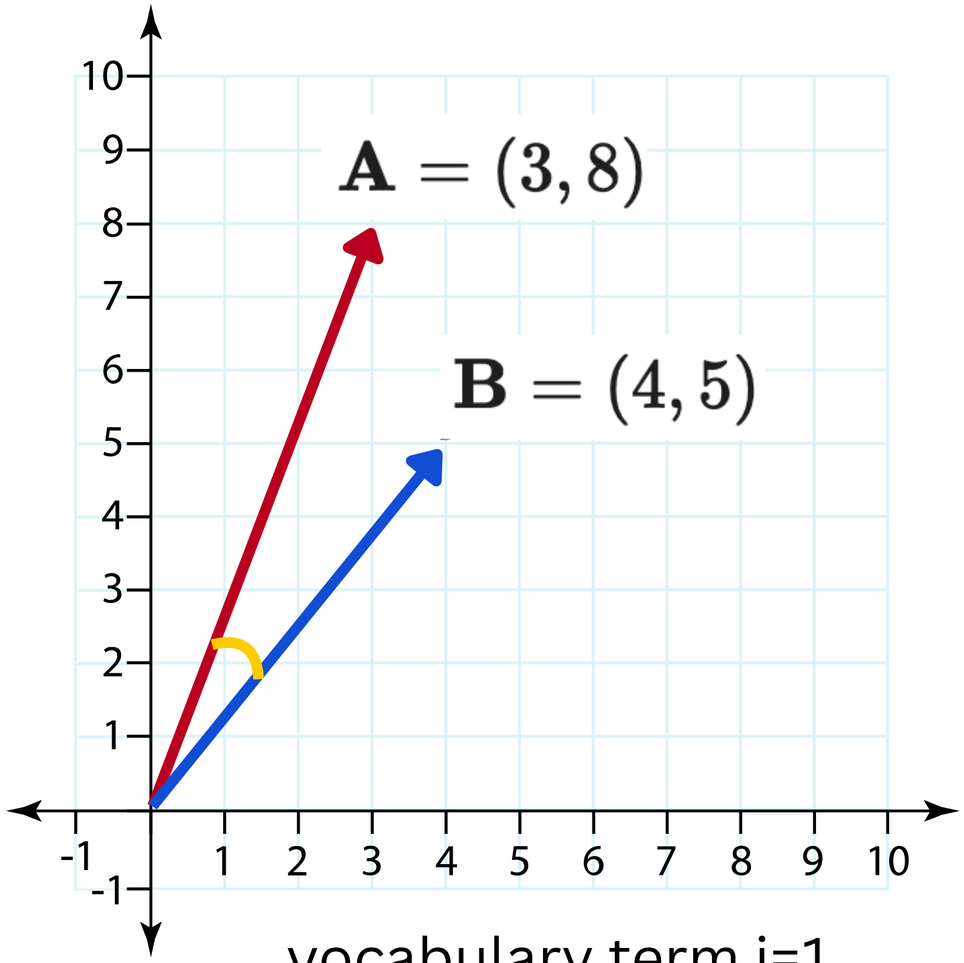


vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{52}{\sqrt{73}\sqrt{4^2 + 5^2}}$$

Cosine Similarity: Math

vocabulary term i=2

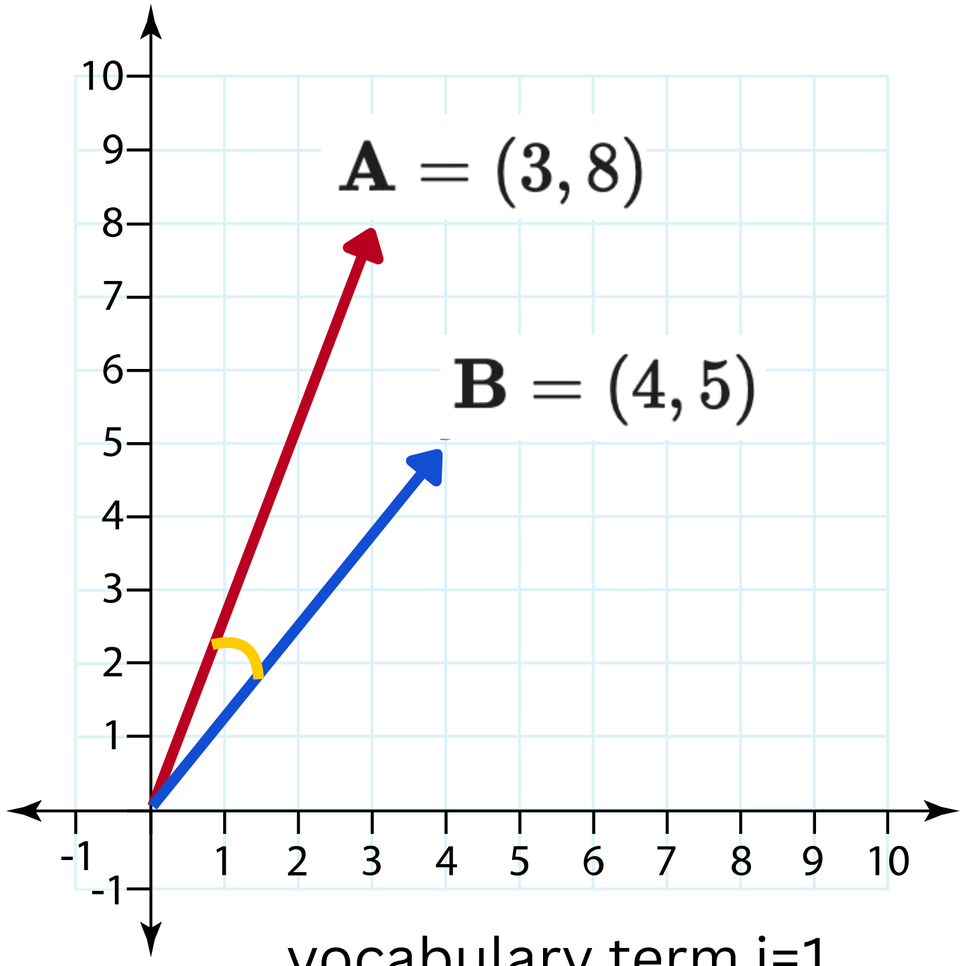


vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{52}{\sqrt{73}\sqrt{41}}$$

Cosine Similarity: Math

vocabulary term i=2

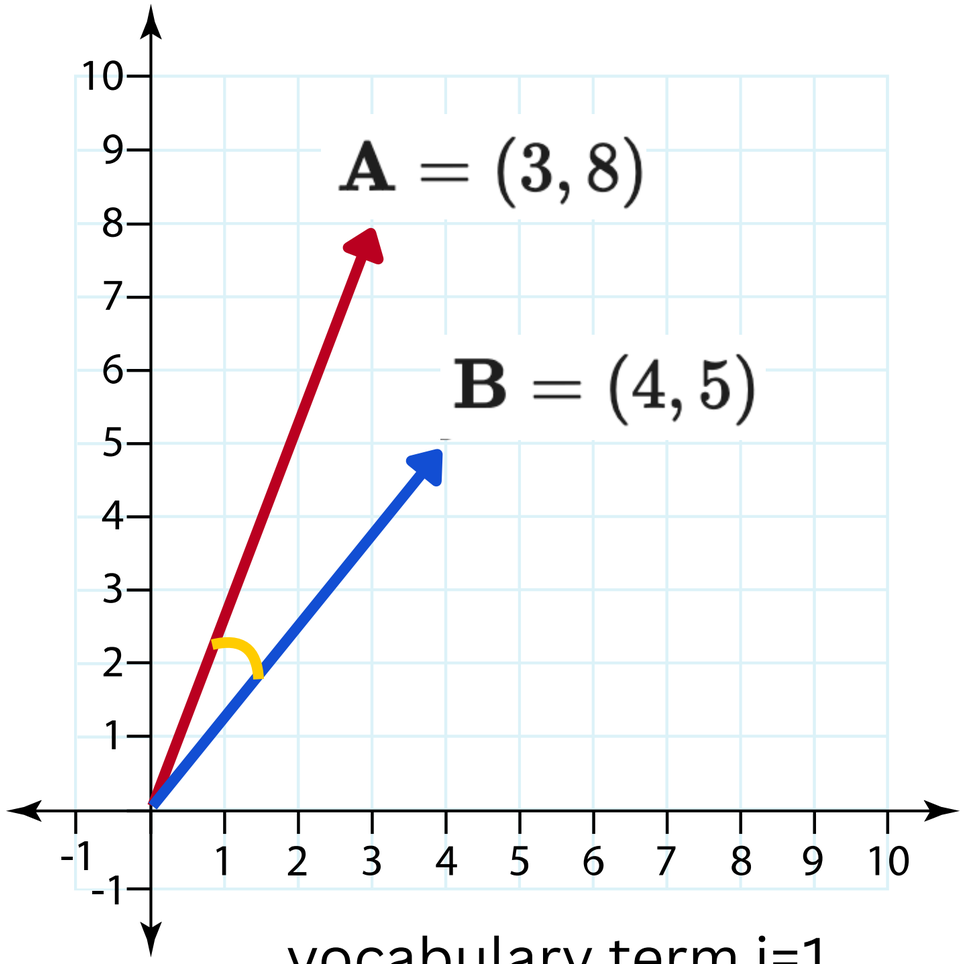


vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{52}{54.708}$$

Cosine Similarity: Math

vocabulary term i=2

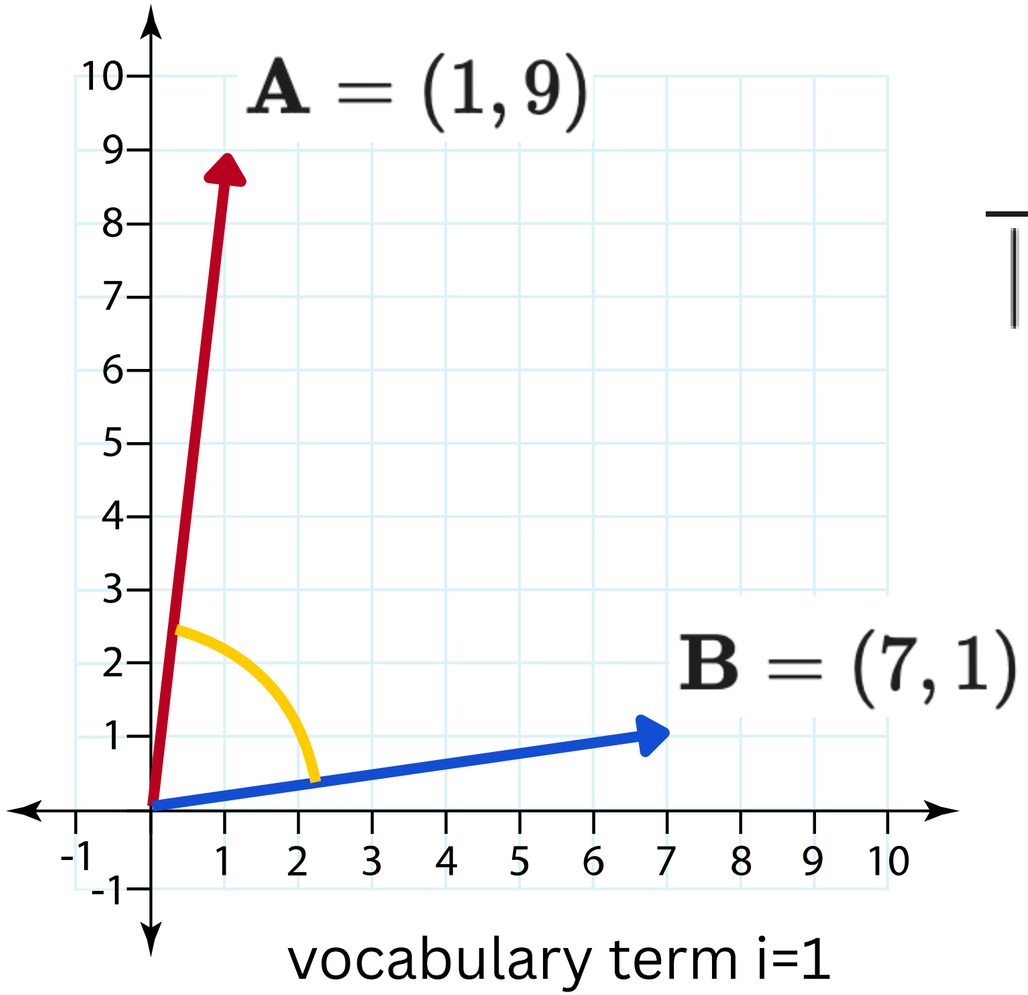


vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = 0.95045$$

Cosine Similarity: Math

vocabulary term i=2

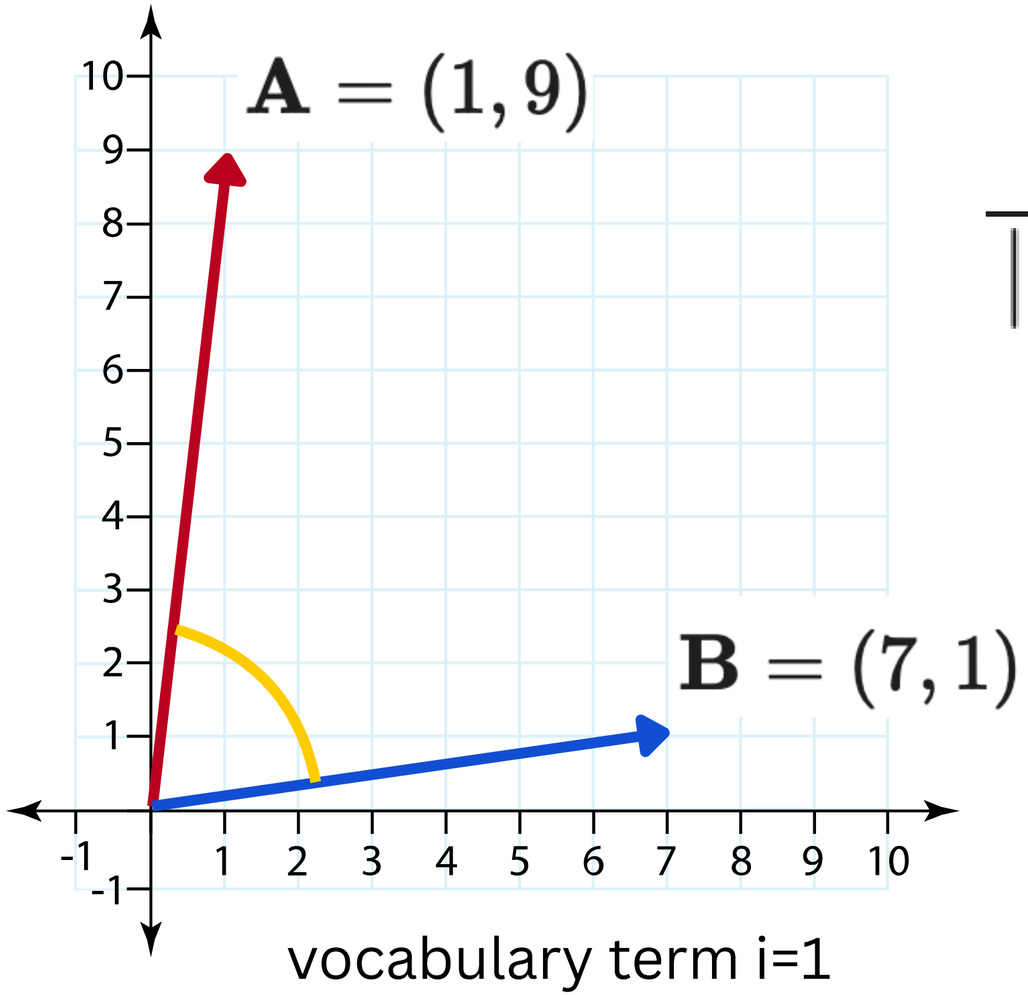


vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{1*7 + 9*1}{\sqrt{1^2+9^2}\sqrt{7^2+1^2}}$$

Cosine Similarity: Math

vocabulary term i=2

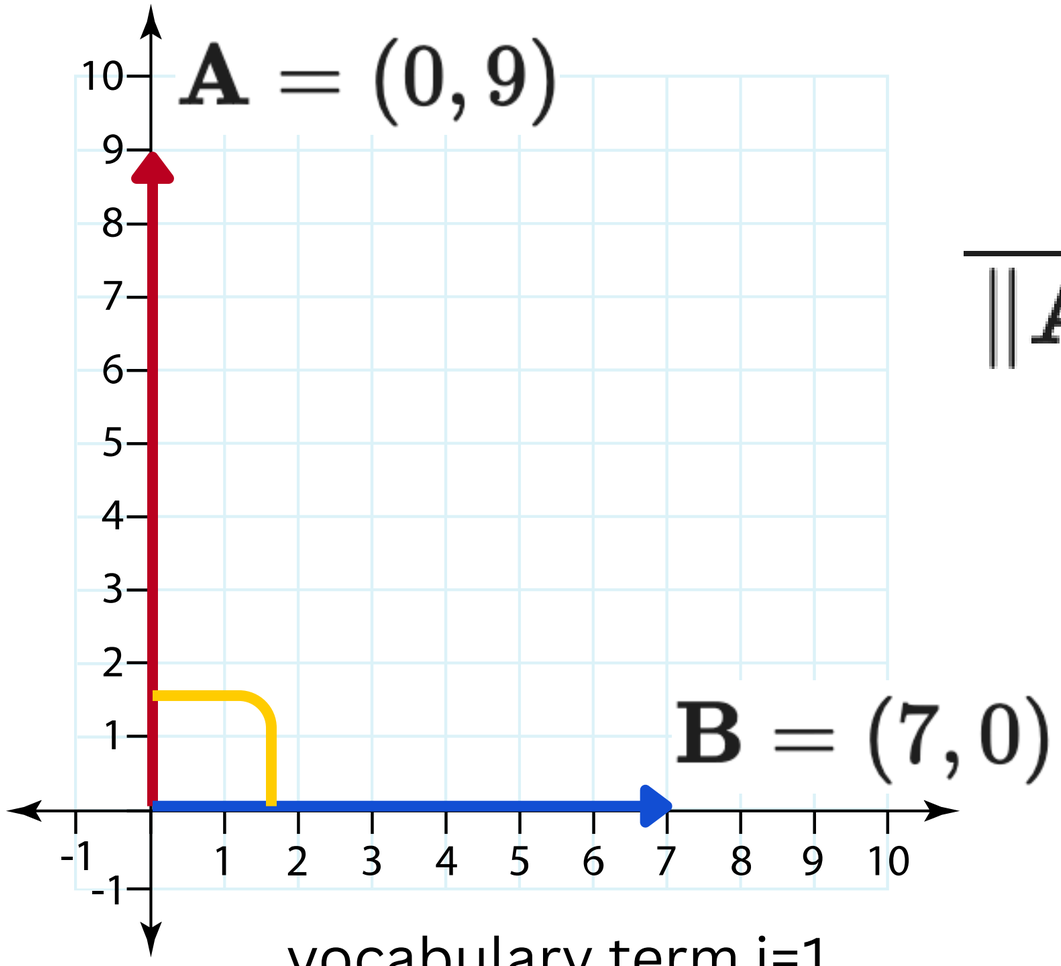


vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = 0.24951$$

Cosine Similarity: Math

vocabulary term i=2

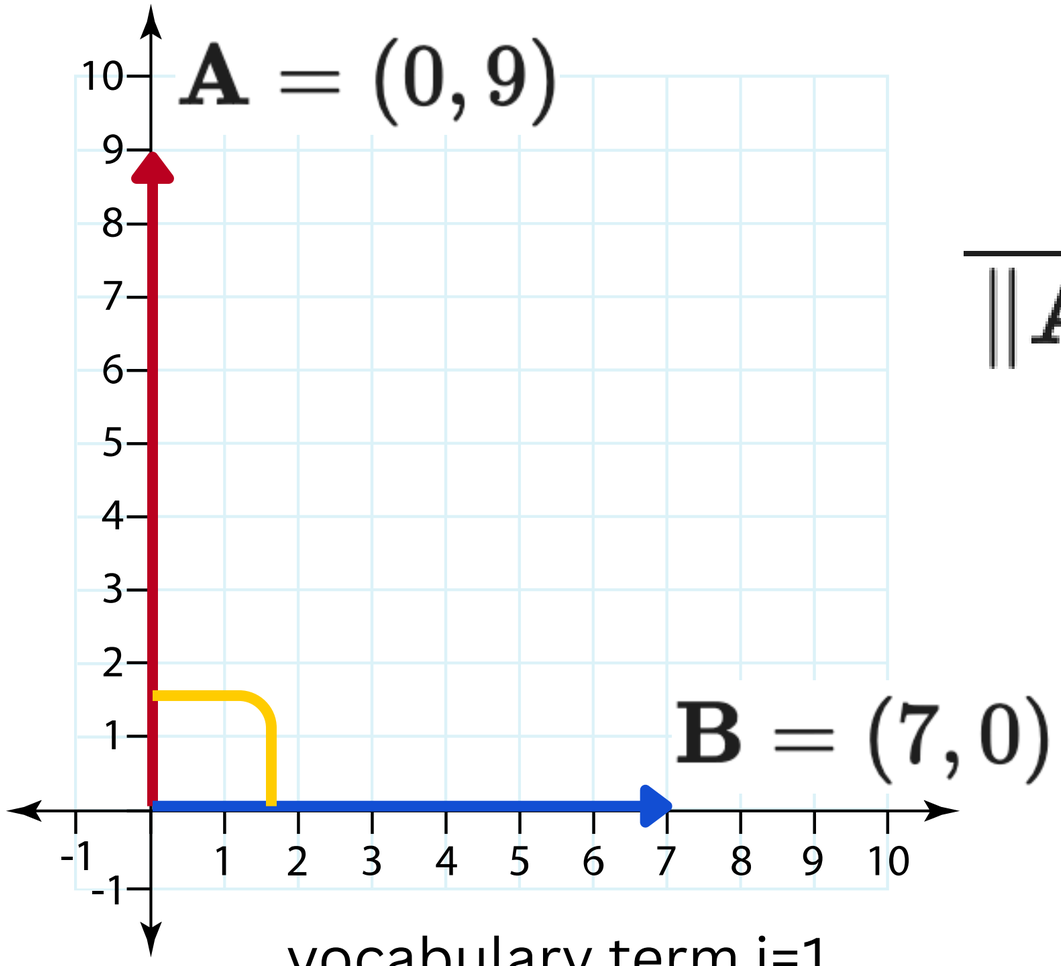


vocabulary term i=1

$$\frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{0*7+9*0}{\sqrt{0^2+9^2}\sqrt{7^2+0^2}}$$

Cosine Similarity: Math

vocabulary term i=2



vocabulary term i=1

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = 0$$

Search Engine

Natural Language Processing Lab 3

</> Code

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1 Objectives

Welcome to the third Natural Language Processing lab!

Today's objectives are to:

1. 🔢 Explore different **text representations** as vectors of token weights
2. 📐 Calculate **cosine similarities** between documents.
3. 🔎 Build a **search engine** to compare a user query with corpus documents.

Next Week: Embeddings! 🤓

